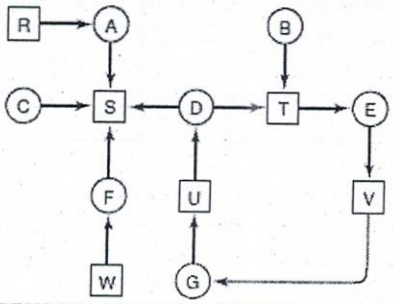
International Islamic University Chittagong

Department of Computer Science and Engineering
B. Sc. in CSE
Final Exam, Autumn 2023

Course Code: **CSE-3631** Course Title: **Operating Systems**
Time: 2 hours 30 minutes Full Marks: 50

		Group A			
1.	a)	Define Deadlock. In the resource allocation graph (RAG) shown in the following figure, is there any deadlock, and if so, which processes are involved? In this graph, the square nodes represent resources, and the circular nodes represent processes. 	CO2	Ap	5
1	b)	In a multi-process system, there are three types of resources (X, Y, and Z) with initial availability (8, 6, 4) units, and five processes (P1, P2, P3,P4,P5) with different maximum needs and current allocations. Using the Banker's algorithm, calculate the maximum need matrix, allocation matrix, and available resources at each step to demonstrate how resources are allocated to the processes while maintaining system safety and preventing potential deadlocks. Provide a clear example of the allocation process.	CO2	Ev	5
		OR			
1.	b)	Consider the following resource allocation state of a system with four processes, P1, P2, P3, and P4, and five types of resources, RS1, RS2, RS3, RS4, and RS5 where C :the current allocation matrix, R: the request matrix. E :the existing resource vector A : the available resource vector	CO2	Ev	5
		1. Using the deadlock detection algorithm, show that there is a deadlock in the system. Identify the processes that are deadlocked. 2. Explain how the system can recover from the deadlock using recovery through preemption.			
2.	a)	What is paging? Explain the concept of free frames in memory management and how they are used for process allocation. Provide an example of a scenario where you have 16 free frames initially. Two processes, Process A and Process B, request memory allocation. Describe the state of free frames before and after the allocation, and discuss the significance of efficient frame allocation in a multi-process environment.	CO3	Un	5

Process	Allocation	Max
P1	7 5 3	2 2 2
P2	3 2 2	2 1 0
P3	9 0 2	3 0 2
P4	2 2 2	0 1 1
P5	6 4 2	4 3 1

C =

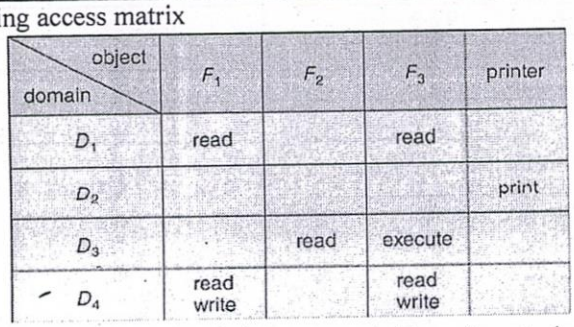
0	1	1	1	2
0	1	0	1	0
0	0	0	0	1
2	1	0	0	0

R =

1	1	0	2	1
0	1	0	2	1
0	2	0	3	1
0	2	1	1	0

E = (24144)

A = (01021)

2.	b)	Discuss the impact of page faults on the overall performance of a computer system and explain how page fault handling mechanisms, such as demand paging, help minimize the effects of page faults.	CO3	Un	5
		OR			
2.	b)	In a computer system, the cache has a Cache Hit Time (H) of 1 nanosecond (ns), a Cache Miss Penalty Time (M) of 10 milliseconds (ms), a Cache Hit Rate (Hr) of 90%, and a Cache Miss Rate (Mr) of 10%. Calculate the Effective Access Time (EAT) for this cache configuration. How does EAT provide insight into the performance of the cache in terms of average memory access time?	CO3	Un	5
		Group B			
3.	a)	Explain the LRU (Least Recently Used) page replacement algorithm in the context of memory management. Using a provided sequence of page references and a memory capacity of 4 pages, demonstrate how the LRU algorithm works to manage pages in memory. Calculate the page fault rate for the given sequence. Discuss the advantages and limitations of the LRU algorithm in comparison to other page replacement algorithms. you have the following page reference string: 2 3 2 1 5 2 4 5 3 2	CO3	An	5
3.	b)	Explain the steps involved in handling a page fault in a virtual memory system. Discuss the key responsibilities of the operating system during this process, including checking page validity, locating the required page, selecting victim pages (if necessary), and updating the page table. Provide an example scenario to illustrate the page fault handling process explain with proper diagram.	CO3	Ap	5
		OR			
3	b)	Illustrate with an example how the Enhanced Second-Chance Page Replacement Algorithm works.	CO3	Ap	5
4.	a)	Explain the different file access methods used in operating systems and file systems. Compare and contrast the characteristics of sequential access, direct access, and indexed access methods. Provide examples of scenarios where each access method is most suitable and discuss the advantages and limitations of each in terms of data retrieval and storage.	CO3	Ap	5
4.	b)	Explain the working procedure of Hashed Page Tables with an example.	CO3	Ap	5
5.	a)	Consider the following access matrix  Now , how can you implement copy rights, owner rights and control rights in the given access matrix?	CO3	Ev	5
5.	b)	Suppose you have two parties, Alice and Bob. Alice wants to send an encrypted message to Bob. They both have their key pairs. Alice's public key (e, n): (7, 77) Alice's private key (d, n): (43, 77) Alice wants to send the message "M" to Bob, but she doesn't want anyone else to read it. The message "M" is represented as a numerical value: M = 9. Using Alice's public and private keys mentioned above, calculate the encrypted value C of the message M = 9 following the RSA encryption process. Then, calculate the decrypted message "M" using Alice's private key.Provide the numerical values for C and M after the encryption and decryption steps.	CO3	Un	5
		OR			
5.	b)	Trojan horse, Spyware, Ransomware and Trap door are the popular malware to breach the security of a system. Create a security plan to protect against these threats.	CO3	Un	5